

ABSTRACT

The present scientific review article aims to explore the complicated role of hormones in the regulation of numerous physiological functions and balance thereof. Serving as special messengers, hormones do not perform any direct functions but transmit orders or directives to other organs, regulating diverse functions, such as metabolism, growth, reproduction, and mood stabilization. This paper will provide insight into insulin, thyroid hormones, and sex hormones and describe their specific mechanisms of action and influence upon the body. Insulin is a pancreatic hormone responsible for blood sugar levels that is released to the body in response to high glucose levels. Thyroid hormones T3 and T4 regulate metabolism, growth, and brain development, particularly intensive during pregnancy. Sex hormones, especially estrogen and progesterone, are responsible for fertility, menstruation, and stable mood. Moreover, the article sheds light on the connection between hormonal imbalances and mental diseases in the form of depression and anxiety. The role of hormones in the HPA axis is accentuated with the example of cortisol. The influence of hormones on addictive behavior is explored further as testosterone and progesterone influence addiction tendency on a separate level, while relationships and high parental behavior depend on them as well.

INTRODUCTION : Hormones

Hormones are chemicals synthesised and produced by specialised glands called endocrine glands to control and regulate the activity of cells and organs in our body, they essentially function as messengers, from the endocrine glands delivered through the bloodstream to certain organs, skin, muscles, and other tissues. These messengers are responsible for controlling many physiological functions (metabolism, growth and development, sexual function, reproduction, sleep-wake cycle, and mood) as well as for maintaining homeostasis in the body which is the constant internal balance, such as blood pressure and blood sugar regulation, fluid (water) and electrolyte balance, and body temperature.

How do hormones play a huge part in regulating such functions?

To understand this we are going to take 2 essential hormones and explain in detail how they function and their effects on the body :

I - Insulin:

What is it? How does it work? How and why it's produced and released?

- Insulin is a crucial hormone produced by the pancreas (an organ that is located behind the stomach) and is important in regulating blood sugar (glucose) levels in the body.

- Production and Release of Insulin:

The hormone Insulin is produced within specialised cells called beta cells, which are located in clusters called the Islets of Langerhans in the pancreas, these cells have glucose-sensing mechanisms that allow them to detect changes in blood glucose concentration. When the

glucose levels rise in the bloodstream, for example after a meal, glucose enters the beta cells through glucose transporters, leading to an increase in the intracellular concentration of ATP (adenosine triphosphate) within the beta cells. This triggers a cascade of events that result in the release of insulin.

- Mechanism of Action:

Once the insulin is released into the bloodstream, it travels to various tissues and organs, including muscle, fat, and liver cells, and then binds to specific receptors on the surface of these target cells, this action initiates a series of several intracellular signals. One of the primary effects of insulin is to promote the translocation of glucose transporters (GLUT4) to the cell membrane. The more GLUT4 transporters were on the cell membrane, the more effective it was for the cells to take up glucose from the bloodstream. Once inside the cells, glucose can be either stored as glycogen (in muscle and liver cells) or utilized for energy production. Once blood glucose levels drop to normal, insulin excretion decreases, preventing hypoglycemia (low blood sugar).

II- Thyroid hormones (T4 and T3)

What are they?

Thyroid hormones are the hormones that are released and produced by the thyroid gland, being triiodothyronine T3 and thyroxine T4, and they are tyrosine-based hormones that they have several physiological effects and functions which include:

- Increases the basal metabolic rate.
- It can induce lipolysis or lipid synthesis depending on the metabolic status.
- Stimulate the metabolism of carbohydrates.
- Anabolism of proteins, induce catabolism of proteins in high doses.
- In children, thyroid hormones act synergistically with growth hormone to stimulate bone growth.
- During the prenatal period, it is needed for the maturation of the brain.
- Thyroid hormone affects fertility, ovulation, and menstruation.

Since thyroid hormones have a lot of functions, let's only focus on how they affect pregnancy.

- Regulation of thyroid hormone during pregnancy:

It involves a complex coordination of hormonal signals. The impact of estrogen is very crucial because when it is released in high doses during pregnancy, it leads to a significant increase in thyroid-binding globulin (TBG) levels, (TBG is a carrier protein that transports thyroid hormones in the bloodstream) therefore it results in a decrease in free thyroxine (T4) concentrations, prompting the pituitary gland to secrete more thyroid-stimulating hormone (TSH). Elevated TSH levels stimulate the thyroid gland to produce and secrete more thyroid hormone, thus ensuring a balance in hormone levels necessary for maternal and fetal health.

- The importance of thyroid hormone in pregnancy:

Thyroid hormones, particularly thyroxine (T4) and triiodothyronine (T3) play a critical role in regulating metabolism and supporting foetal brain development. Proper levels of thyroid hormone are essential for neuronal migration, differentiation, and synaptogenesis, influencing

cognitive and motor function in the developing fetus. The demand for thyroid hormones increases significantly during pregnancy, with the peak demand typically occurring around the 20th week and persisting until birth. Maternal thyroid hormones also serve as a vital source for fetal thyroid function, providing essential support for proper growth and development.

- Thyroid hormone release during pregnancy:

It is regulated by various factors, including the stimulation initiated by β -HCG hormone in the first 3 months of pregnancy. This hormone shares structural similarities with thyroid-stimulating hormone (TSH) and overrides the normal feedback system. Additionally, changes in iodine metabolism and the active transport of maternal thyroid hormones across the placenta contribute to the release and regulation of thyroid hormones during pregnancy. These mechanisms ensure that both the mother and fetus receive adequate thyroid hormone levels necessary for optimal health and development throughout pregnancy.

Menstrual Cycle

The interplay between hormones should be well-regulated during the menstrual cycle. From menstruation to the luteal phase, the “female” hormones are expected to be working together as a whole single instrument. They affect certain things like moods and energy levels. Fluctuations in a woman's hormone levels are normal and necessary, but sometimes they can go out of balance, and that leads to results that are preferred to be avoided.

But what are these ‘Female hormones’? And how do they negatively affect the woman’s body during the menstrual cycle when imbalanced?

Oestrogen & Progesterone – Production and Functions :

Ovaries are where estrogen is mainly produced, however, it is also produced in the adrenal glands and fatty tissue in our body. Everyone needs estrogen, not only people who have periods. However, in those who do, it is present at much higher levels, and it has a significant role in regulating fertility and the menstrual cycle. Oestrogen also has other functions in the health of other organs such as the heart, the brain, the pelvic muscles, etc. Oestrogen levels rise during puberty in young teens to help their body to mature and develop. When a woman is during her menstruating years, to trigger ovulation and increase blood flow to the vagina, oestrogen levels increase and are at their highest levels mid-cycle. Then on periods, they reach their lowest levels.

Preparing the endometrium is what the oestrogene focuses mainly on, for the expected implantation and development of a fertilised egg. During the menstrual cycle, and specifically after ovulation, progesterone levels rise, thickening the uterine lining to create the best home for embryo implantation. If the egg does not get fertilised, progesterone levels decrease, which results in the shedding of the uterine lining during menstruation. Otherwise,

progesterone continues to rise, enhancing the thickened uterine lining and providing nutrients for the growth of the embryo. Throughout pregnancy, progesterone levels reach their peak in the third month and drop gradually leading up to menopause as ovulation ends.

Estrogen – Imbalance and Memory

It has been scientifically proven that estrogen affects the memory function in the brain with its neuroprotective and neurotrophic effects; the blood flow increases thanks to the estrogen hormone to supply the neurons with glucose and oxygen, it increases neurotransmitter levels and decreases the production of Alzheimer-linked beta-amyloid peptides neurons. Therefore several studies state that an imbalance in estrogen levels during normal aging may promote the risk for Alzheimer's disease (a type of dementia), despite that average levels of estrogen did prevent the development of the disease, once symptoms start occurring, it did not appear to slow or reverse the process of it. The testers who used estrogen supplements during or a bit after menopause were less likely to develop Alzheimer's disease than those who had not been on this treatment.

It's been studied several times the cognitive function in young women and how different E2 (estradiol, also spelled oestradiol, is an oestrogen steroid hormone) levels could play a role in affecting it during the menstrual cycle. They reached a conclusion that states that high E2 levels are linked with increased memory, verbal fluency, verbal declarative memory, articulation, and working memory. On the contrary, low E2 levels are associated with increased visuospatial abilities.

In addition, oestrogen regulates learning and memory in a dissociable manner based on attributes of the behavioural task and neural systems engaged. For example, increased levels of estradiol, enhance performance on spatially-driven, working memory, or explicit tasks that rely on the hippocampus, such as the radial arm maze, swim task, delayed matching-to-position, place task, and object placement.

Oestrogen & Progesterone - Depression

It has been indicated that gonadal hormones heavily affect mood in women. Depression and anxiety are common struggles for women during times of significant hormonal changes; including the premenstrual, postpartum, and perimenopausal periods. It is crucial to mention that depression and anxiety are complex, multifactorial disorders, and not the hormones alone influence the patient's brain chemistry, so naturally, other major factors contribute to the hormonal impact on mood; such as age and the dosage of any chemical treatment taken by women, genetics, socio-cultural roles, and importantly, environmental stressors.

The symptoms of depression are visible during women's transition periods; like menopause, premenopausal, pregnancy or post-delivery etc. Due to fluctuation of the hormonal balance. Piling up these conditions can lead to clinical depression. Since the main reason behind this mental issue is hormonal imbalance, the best-suited therapy for treating these critical periods is hormonal therapy (oestrogen and transdermal estradiol). When women are in their pregnancy and breastfeeding phases, they must be cautious of antidepressant medication.

The studies strongly suggest that both increased stress sensitivity and constant fluctuations of ovarian hormones throughout the reproductive cycle lead to periodic loss of the protective effects of oestrogen, resulting in women suffering from depression and anxiety disorders. Given that estrogens positively modulate important transmitter systems modulating mood and reward (i.e., serotonin, dopamine), periodic E2 withdrawal consequent to stress, parturition, and even the normal cycle may make women more sensitive to stress-related psychopathologies.

Oestrogen is needed in sufficient concentration to increase and maintain the levels of serotonin, a neurotransmitter that can prevent mood change, irritability, and depression. Low oestrogen levels may lead to a decrease in serotonin concentration and this dysregulation of postsynaptic serotonergic function results in depression, anxiety and sleeplessness. An important element in the serotonergic system, 5-HT receptors, has been proven that manipulation of oestrogen levels affects its functions. Oestrogen helps in the increase of serotonin levels by altering mRNA and protein levels of various serotonin markers, decreasing serotonin breakdown. Progesterone affects GABA (gamma amino-butyric acid) receptors in the brain, when GABA levels decrease, after a stress-induced neurotransmitter release, our body cannot easily relax. These factors have significant impact on female mood disorders such as premenstrual disorders and depression during pregnancy, postpartum, and during the menopausal transition.

It is required to do more detailed studies in order to understand and overcome other related factors which could possibly elevate the symptoms. As a strong concern over women's health issues is arising, it will be a relief to spread awareness on women's health and also obtain medical therapy for treatments.

Hormonal imbalance – Postpartum OCD and intrusive thoughts

Only recently there has been rising interest in OCD in the perinatal period, several studies and reports have stated that pregnancy and the puerperium may expedite or trigger OCD in some women. Postpartum OCD (ppOCD) presents distinct features, with contamination fears and unwanted thoughts of harming one's infant being prevalent.

Postpartum Obsessive-Compulsive Disorder (PPOCD) is an overlooked significant struggle affecting approximately 1 in 18 new mothers right after giving birth. While it's common and

normal for new mothers to experience various mood fluctuations and stress postpartum, PPOCD stands as a more severe condition, concerningly impacting a mother's ability to take care of her own well-being and that of her newborn. Unlike the more widely recognized postpartum depression (PPD), the existence of Postpartum OCD is still unknown to many and never heard of, despite the fact of its harmful impact on new mothers.

What's peculiar about PPOCD are intrusive thoughts, which can be triggering and unsettling for new mothers. These intrusive thoughts are often on and about being terrified of unintentionally harming the baby, doubting oneself to be a mother, or overwhelming feelings of guilt and sadness. During pregnancy and till postpartum, it is believed that hormonal changes outrage these uncomfortable thoughts, contributing dramatically to the development and exacerbation of PPOCD and its associated symptoms.

Furthermore, evidence coming to light focus on the complex relationship between reproductive phases and the severity of Obsessive-Compulsive Disorder (OCD). A recent study conducted with 350 female outpatients with OCD explored the impact of menstrual cycle periods, pregnancy, menopause, hormonal contraceptives, and selective serotonin reuptake inhibitors on OCD symptom severity. Remarkably, a significant portion of participants reported increased symptoms during the premenstrual period, menopause, and pregnancy, with some experiencing relief during pregnancy. These reports emphasize the continuous effect of hormonal fluctuations on the worsening and exacerbation of OCD symptoms, specifically during critical reproductive phases.

Additionally, the perinatal period, starting from pregnancy through the first year after delivery, stands out as a sensitive time for obsessive-compulsive and its symptoms. Changes in hormones levels, especially fluctuations in estrogen levels, play a crucial role in influencing OCD symptoms during this phase. The symptoms may exacerbate due to the increased estrogen levels for some, the drop in estrogen levels during childbirth and the postpartum period may result in others being more susceptible to OCD. The complicated interplay between estrogen and various neurotransmitter systems, including serotonin, dopamine, and glutamate, is believed to be contributory in understanding the intricate etiology of perinatal OCD. Moreover, while research on the roles of progesterone, allopregnanolone, and oxytocin in perinatal OCD remains limited, latest studies states their potential importance in influencing the nature of this condition.

Hormone imbalance – Polycystic Ovarian Syndrome (PCOS)

PCOS is an endocrine disorder that is, in most cases, the main reason for other menstrual disorders characterized by chronic anovulation, hyperandrogenism, and infertility along with various metabolic disorders. In around 10% to 13% of all women of childbearing age report infertility and anovulation due to PCOS. have also been reported due to PCOS. Some women

have polycystic ovaries while some may have single ovarian cysts. Insulin resistance is considered as the main cause of PCOS (as explained in the insulin section, that is responsible for the ability of the body to break down and absorb glucose). It is known that the ovaries are sensitive to insulin, therefore, when its levels are imbalanced, ovaries try to recompense for it by producing more ovarian hormones which results in disturbance in several metabolic processes of the body. Some symptoms may include acne, infertility, facial hair growth, head hair loss, and dark patches of skin on the neck, or skin tags. Other reported causes are elevated testosterone concentrations and any mutation in Luteinizing Hormone and/or its receptor.

A high Gonadotropin hormone-releasing hormone GnRH release frequency will lead to an elevated release of Luteinizing hormone LH and a decreased release of Follicle-stimulating hormone FSH and, in many patients, the producing cells of LH react heavily to GnRH resulting in an extreme LH production. As high concentrations of LH and a low FSH are released, the ovarian follicles will not develop, which lead to lack of estrogen production. In the follicles there are cells responsible for producing estrogen are the follicular cells and the thecal cells, they first produce androgens, that will be used to produce estrogens. Because of the low levels of FSH however, it will lead to a lack of granulosa cells, eventually to hyperandrogenism. Moreover, high levels of reduced androgens lead to further inhibition of aromatase activity, thereby reducing estrogen production even more. All of these irregular release and production of hormones will lead to very high levels of Anti-Müllerian hormone AMH. The increased AMH levels will lead to further inhibition of FSH release. The high levels of androgens and the elevated insulin levels due to insulin resistance will lead to a decrease in the production of Sex Hormone Binding Globulin (SHBG) which, as a result, leads to increased levels of androgens in the blood, and thus will exacerbate the effects of androgens. Since the effects of androgens are enhanced, insulin resistance will be increased. High androgen levels in obese women suffering from PCOS, will lead to the production of estrogens in the adipose tissue due to the presence of aromatase in the adipocytes.

Hormone imbalance and other mental illnesses

Life stress exposure plays a significant role in the pathophysiology of both depression and anxiety. Of note, dysregulation of cortisol, a key stress hormone, is linked with the onset and/or severity of a variety of affective diseases, including bipolar disorder, anxiety, post-traumatic stress disorder and major depression. Glucocorticoids are the principle output of the hypothalamo-pituitary-adrenocortical axis, a system integral to maintenance of homeostasis. Activation of the HPA axis by external stimuli (e.g., stress) triggers a series of neuroendocrine responses. Real or perceived stressors stimulate hypothalamic neurons in the medial parvocellular PVN. Once activated these neurons synthesize and secrete ACTH secretagogues (corticotrophin releasing hormone (CRH) and arginine vasopressin (AVP)), which stimulate pituitary corticotrophs, causing release of adrenocorticotrophic hormone (ACTH) into systemic circulation. ACTH then stimulates the adrenal cortex, resulting in the synthesis and release of glucocorticoids (cortisol in humans, corticosterone in rats).

Glucocorticoids, the end product of this system, exert negative feedback at the level of the pituitary, hypothalamus and extra-hypothalamic sites including the hippocampus and medial prefrontal cortex, thereby limiting prolonged activation of this system. Elevated glucocorticoids seen in depression are believed to result from deficient negative feedback, resulting in increased glucocorticoid signaling at the level of the brain. It is thought that central glucocorticoids promote depression-like behavior, triggering clinical trials of the antidepressant efficacy of glucocorticoid receptor antagonists.

It has been proven by several studies that dysregulation of cortisol, a key stress hormone, is linked with the onset and the severity of a variety of affective diseases, including bipolar disorder, anxiety, post-traumatic stress disorder, and major depression. Glucocorticoids are hormones that are mainly produced by the hypothalamic-pituitary-adrenal (HPA) axis, which is a system integral to the maintenance of homeostasis (balance in the body). An external stimuli can trigger and activate the HPA axis that is per example, stress, whether real or perceived. This starts with series of neuroendocrine responses. Once stimulated, these neurons release chemical substances (like corticotrophin releasing hormone CRH and arginine vasopressin AVP), which command the pituitary gland to release adrenocorticotrophic hormone (ACTH) into systemic circulation. After that ACTH signals the adrenal glands to produce and release glucocorticoids (cortisol in humans), this is a way of our body to cope with stress. When enough glucocorticoids are made, it is reported to the brain by sending signals back to it to slow down the process of releasing CRH and AVP, that is called negative feedback. In certain scenarios like depression, this feedback mechanism might not work accordingly, resulting in high concentration of glucocorticoid signaling in the brain, which is believed to the lead of depression-like behavior.

Hormones and addiction

Neurohypophyseal hormones:

Several researches has been conducted on animals report that certain hormones in the brain, such as oxytocin and vasopressin, interplay in the course of addiction. For example, there has been tests on rats dependent on substances like nicotine or heroin, giving them oxytocin, which resulted in reduction of the withdrawal symptoms. Oxytocin seems to have an impact on different parts of the brain involved in addiction, like the reward circuit and areas responsible for feelings of dependence. It may as well affect the dosage of such substances; alcohol and cocaine, a person yearns to abuse. These studies state that oxytocin hormone could be a potential therapy for addiction, and must be explored further by clinical trials.

Stress hormones:

Psychotropic drugs, which influence the mind's chemistry, have been found to trigger the release of stress hormones in the body. In studies conducted with adult male Sprague-Dawley rats, a single injection of alcohol led to the activation of the Hypothalamic-pituitary-adrenal (HPA) axis, that is being the body's stress response system. With acute alcohol consumption, this response is noticeably common. However, in cases of chronic alcohol abuse, there is a developed tolerance to this stress response, which can eventually lead to an increased risk of relapse and heavy drinking. Neuroadaptation is a mechanism underlying relapse, during chronic drug use, this process is characterized by compensatory changes in the brain's response. These changes usually result in elevated levels of stress hormones in chronic drug abusers, predisposing them to fall into drug-seeking behavior. Studies conducted with substances like alcohol and cocaine in rodents have shed light on this phenomenon.

Sex hormones:

Recent studies state that sex hormones are responsible for determining the period length and of an individual's addiction, and the amount consumed of the substances by them. In animal researches, male rats are more likely to develop an alcohol addiction faster than female rats, and experience slower recovery compared to females. However, females seem to consume more alcohol, specifically during adolescence and adulthood. These gender differences in alcohol consumption emphasize the impact of sex hormones on behaviors related to addiction. Additionally, sex hormones can affect several stages of addiction such as acquisition, maintenance, dysregulation-escalation, and relapse. Higher levels of stress reactivity are noticeable to be experienced by women, leading to difficulty in quitting addictive substances compared to men.

Summary:

In summary, this review sheds light on the critical functions of hormones in regulating various physiological systems and preserving homeostasis. From insulin to thyroid hormones, and from the sex hormones, numerous systems and pathophysiological processes have a role in metabolic, growth, reproductive, mood and emotion-related pathologies. As depicted in this review, the impairment of hormone functions is involved in numerous disorders ranging from metabolic to infertility issues and depression and anxiety. Hormones also contribute to addictive behaviors in susceptibility and the mediating effects of sex hormones. Moreover, oxytocin has potential in the therapeutic context as well. By elucidating the complex interplay between hormones and various physiological processes, this article underscores the importance of hormonal balance in promoting overall health and well-being.

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